Pawan Prasad Padhy

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LinkedIn | GitHub | Scaler | Portfolio

SKILLS

- **Programming Languages:** Python, HTML, CSS, JavaScript
- **Libraries and Frameworks:** Django, FastAPI, Flask, Python Libraries, React, Node.js, Express.js
- **Databases:** MongoDB, MySQL, PostgreSQL, SQL, NoSQL
- **Cloud & DevOps:** AWS, Git, GitHub, GitHub Actions, Linux/Unix, Airflow, Datadog, Grafana
- **Tools:** Postman, REST API, TensorFlow, NumPy, Pandas
- **Data Visualization:** Tableau, Power BI, Apache Superset
- **Currently Learning**: *Apache Tools (Spark, Hadoop, Hive), Kafka, Data Modeling Tools, Data Warehouses (Redshift, BigQuery), Snowflake*

EXPERIENCE

Software Engineer Nov 2025 - Present
Recro (Client: Allen Digital) Bangalore, Karnataka, India

- Designed and orchestrated Airflow DAGs implementing ETL/ELT workflows for exam data ingestion and processing, improving pipeline monitoring and reliability across the exam operations stack.
- Operated a multi-system data architecture spanning MongoDB (EC2-hosted and Atlas), ClickHouse (analytical / OLAP store), and S3, performing real-time data synchronization across three-distinct storage layers to support high-scale exam operations.
- Built observability and business-facing dashboards using Grafana, Superset, and Power BI/Tableau (as needed) to visualize exam state transitions, system performance and operational metrics for cross-functional stakeholders.
- Automated exam state transition logic to detect and resolve exam IDs stuck across multiple failure states, reducing manual intervention in the data pipeline and improving end-to-end pipeline reliability.
- Managed and deployed application configurations across AWS environments, standardizing infrastructure setup to ensure consistent, repeatable deployment of data pipeline components.
- Architected a dual-pipeline concurrent processing system in Python (asyncio), handling 20+ parallel bot-solution generations and LaTeX-to-image conversions for thousands of records, with automatic retry on failure - applying distributed-workload principles at the application level.
- Engineered a data transformation pipeline converting question paper PDFs into structured HTML, mapping subject-specific question IDs into the question bank's data model.
- Automated deployment and scheduling of recurring data-processing jobs on AWS EC2, standardizing operational workflows for consistency and repeatability.
- Contributed to and maintained backend services across multiple Go repositories, diagnosing and fixing bugs alongside ongoing Python-based development work.
- Collaborated with product and operations teams to translate business questions about exam delivery issues into data queries and dashboard views, supporting faster decision-making.
- Used Groundcover for pipeline observability and logging, accelerating root-cause identification for data/production issues.

Software Developer Engineer Nov 2024 - Nov 2025
MountBlue Technologies Pvt. Ltd. Bangalore, Karnataka, India

- Designed normalized PostgreSQL schemas and indexing strategies for document metadata storage, applying database optimization principles to improve query performance and support compliance-driven data archiving.
- Developed and tested web applications, APIs, and automation scripts using Django and FastAPI, establishing CI/CD pipelines with GitHub Actions across multiple repositories.
- Built 'Rex,' a Python-based Playwright automation agent to order client documents across three external portals (Community Archives, Homewise, Ciranet), eliminating manual, repetitive ordering workflows.
- Designed an 'Ambiguity Process' module within Rex to manage document states end-to-end for address-specific cases, automatically retrieving documents alongside delivery dates and pricing.
- Built 'Agent AutoTag,' an LLM-powered agentic system that classifies client documents by orchestrating prompt design, function/tool calling and retrieval-augmented generation (RAG) against a MySQL-backed knowledge store, cutting manual classification efforts across hundreds of documents.
- Engineered the agent's decision pipeline to plan and sequence multiple tool calls - retrieval, tagging logic and definition lookups - against pre-defined prompts, keywords and business rules, integrating outputs into a coherent end-to-end workflows.
- Orchestrated OCR output, MySQL storage, and Celery-based background processing into a single automated pipeline, enabling the agent to tag unstructured client documents asynchronously with minimal manual review.
- Built and maintained CI/CD pipelines for the AutoTag and Rex services using GitHub Actions, automating testing, deployment, and code quality checks while implementing structured logging, comprehensive documentation, and maintainable code to support scalability and seamless handovers.
- Tested and validated RESTful API endpoints using Postman, building comprehensive test collections for the 'Rex' and 'AutoTag' services to strengthen error handling coverage.

Civil Engineer

- Grasim Industries Pvt. Ltd. — Feb 2022 - Sep 2022
- Er. Surendra Kumar Panigrahi (A Class Contractor) — July 2019 - Oct 2021

PROJECTS

Portfolio Website — Live Demo

- Built a full-stack portfolio application using the MERN stack, with a responsive React frontend, dynamic routing, and RESTful Express.js APIs for contact form submissions and content management, backed by MongoDB for persistent storage.
- *Tech Stack (MERN): React.js, Node.js, Express.js, MongoDB, JavaScript, HTML5, CSS3, REST APIs.*

Resume Creator — GitHub

- Built a markdown-based resume editor with a live preview and one-click ATS-friendly PDF export, using a React (Vite) frontend and a Flask backend with ReportLab for server-side PDF generation.
- Implemented a shared markdown parser on both client and server so the browser preview stays in sync with the generated PDF, including bold/italic formatting, links, and multi-page layout.
- *Tech Stack: React, Vite, Flask, ReportLab, Python, JavaScript, Markdown.*

Olist E-Commerce Analytics — GitHub

- Built a full analytics pipeline on the Olist Brazilian e-commerce dataset.
- Cleaned and modelled about 100k orders in SQLite, answered 14 business questions with SQL (revenue trends, category Pareto, delivery delay vs reviews, payment mix, cohort retention), then segmented customers with RFM and published the results in Tableau and Power BI.
- *Tech Stack: Python, Pandas, SQL, SQLite, Tableau, Power BI.*

Trello Clone Application — Live Demo

- Built a Trello clone supporting full CRUD operations across boards, lists, cards, and checklists, using a modular JavaScript architecture with dynamic state management, event delegation, and efficient DOM manipulation without any frontend framework.
- *Tech Stack: HTML5, CSS3, Vanilla JavaScript, Trello REST API.*

IMDB Clone — Live Demo

- Built a full-stack IMDB clone using Django and PostgreSQL, integrating the TMDB API for real-time trending movie data, with genre-based filtering, pagination, and a normalized, indexed database schema.
- *Tech Stack: Django, Python, PostgreSQL, HTML5, Tailwind CSS, JavaScript, TMDB API.*

Image Classification Model - Transfer Learning Pipeline — GitHub

- Built an image classification model using TensorFlow/Keras (MobileNetV2), applying data preprocessing and augmentation techniques on a ~17K-image, 6-class dataset (Intel Image Classification) to improve model generalization.
- Conducted experiments varying augmentation strategies and training configurations to evaluate and improve model robustness.
- *Tech Stack: Python, TensorFlow, Keras, OpenCV.*

CERTIFICATIONS

- Data Engineering on AWS – Foundations (AWS)
- Foundations of Prompt Engineering (AWS)
- Fundamentals of Data Analytics (AWS)
- Tableau Basics (upGrad)
- Certified Summer Course on AutoCAD
- Certified Summer Course on StaadPRO
- Vocational Training at Indian Rare Earth's Limited

EDUCATION

Scaler — 2022 - 2024

Specialisation in Software Development

- **Coursework**: Data Structures & Algorithms, Object-Oriented Design, SQL, System Design, Full Stack Development.
- **Engineering Practices**: Software Design & Documentation, Testing & Debugging, Cross-Functional Collaboration, Multi-Platform Systems (Linux/Unix, AWS, distributed databases).
- Sustained a 1000+ day consistent DSA and coding practice streak, reflecting strong problem-solving discipline and command over core CS fundamentals.

National Institute Of Science And Technology, Odisha — 2015 - 2019

Bachelor of Technology — Civil Engineering